(47)

$a:b:c:d:e = 2\times4\times6\times8 : 3\times4\times6\times8 : 3\times5\times6\times8 : 3\times5\times7\times8 : 3\times5\times7\times9$

$= 128 : 192 : 240 : 280 : 315$

when in question a:c:e asks $\Rightarrow a:c:e = 128:240:315$

Ex:3:- A:B:C:D = 1:2:4:8, 1500 Rs devided among A, B, C & D in the given ratio, then their individual shares are?

Ans:- $(1+2+4+8)x = 1500 \Rightarrow x = 100$

$A = 100, B = 200, C = 400, D = 800$

Ex:-4:- 1500 devided among A, B, C & D in such a way that $A = 2B = 4C = 8D$ then their shares are?

Ans:- Convert all the values in terms of a single unknown.

$A = 8D, \Rightarrow B = \frac{8D}{2} = 4D, \quad C = \frac{8D}{4} = 2D, \quad D = D$

$A:B:C:D = 8D:4D:2D:D = 8:4:2:1.$

$(8+4+2+1)x = 1500 \Rightarrow x = 100 \Rightarrow$ shares $\Rightarrow$ 800, 400, 200, 100 (A B C D)

Ex:-5:- A sum of money devided among A, B, C & D in such a way that $\frac{1}{2}A = \frac{2}{3}B = \frac{3}{4}C = \frac{4}{5}D$ then their ratio is?

Ans:- When fractional ratio's are given to reduce complexity assume all values equals to a single variable 'k'.

$\frac{1}{2}A = \frac{2}{3}B = \frac{3}{4}C = \frac{4}{5}D = k.$

$A = 2k, \quad B = \frac{3}{2}k, \quad C = \frac{4}{3}k, \quad D = \frac{5}{4}k.$

$A:B:C:D = 2k : \frac{3}{2}k : \frac{4}{3}k : \frac{5}{4}k = \frac{2}{1} : \frac{3}{2} : \frac{4}{3} : \frac{5}{4}$

multiple with LCM(denominators) = LCM(2,3,4) = 12

$A:B:C:D = \left[2 : \frac{3}{2} : \frac{4}{3} : \frac{5}{4}\right] \times 12 = 24 : 18 : 16 : 15$

(98)

from the above question solution we can conclude.

5* when ratios are given in such a way one is related with other quantities like 1st one is half of 2nd like. then, their ratios is inverse of ratios of factors given to represent relation b/w them.

$\frac{1}{2}a = \frac{2}{3}b = \frac{3}{4}c = \frac{4}{5}D \Rightarrow a:b:c:d = \frac{1}{1/2} : \frac{1}{2/3} : \frac{1}{3/4} : \frac{1}{4/5}$

$a:b:c:d = \left[2 : \frac{3}{2} : \frac{4}{3} : \frac{5}{4}\right]12 = 24:18:16:15$

Ex:-6:- 30 ltrs of mixture of milk & water containing them in the ratio of 2:3. then.

(i) How many ltrs of water is to be added to make the ratio as 1:4 ?

(ii) half of mixture is replaced with pure water, this process is repeated 2 more times, then the resultant ratio is ?

Ans:- (i) M-1:-

30 → 2 : 3

1 Part = $\frac{30}{5}$ = 6 l

2×6 = 12 lt : 3×6 = 18 lt] +30 l.

1 : 4

1P = 12 l : 4×12 l = 48 l

[We are adding water means milk content is same in the transactions]

30 ltrs of water to be added to make 1:4.

(or)

M-2 $\frac{18+x}{30+x} = \frac{4}{5}$ [water fraction adding water]

$90+5x = 120+4x \Rightarrow x = 30$ ltrs

(or)

M-3 Equalize constant quantity no. of factors in both the ratios.

M $\begin{bmatrix} 2:3 \\ 1:4 \\ 2:8 \end{bmatrix}$ 2 +5P

1P = 6 ltrs

+5P = 5×6 = 30 ltrs of water to be added.